CloudSage — APIs, Data and Contracts

AWS integrations, backend APIs and core data structures

1. AWS Integrations

API/Service	Initial use
EC2	Resource discovery and configuration
CloudWatch	Utilization and health metrics
Cost Explorer	Cost/trend data
STS	Identity/account verification
Terraform CLI/JSON	IaC state/plan/change workflow

2. Later AWS Integrations

- EBS
- S3
- RDS
- ELBv2
- Auto Scaling
- ECS
- Lambda
- EC2 networking
- Cost Allocation Tags
- Budgets where useful

3. Backend API

GET /health
GET /resources
GET /resources/{resource_id}
POST /scans
GET /scans/{scan_id}
POST /analysis/{resource_id}
GET /recommendations
GET /recommendations/{id}
POST /recommendations/{id}/approve
POST /recommendations/{id}/reject
POST /terraform/{id}/plan
POST /terraform/{id}/apply
GET /optimizations/{id}/verification
GET /audit

4. Core Database Entities

Entity	Purpose
resources	Canonical AWS resources
resource_metrics	Aggregated utilization/health
cost_records	Cost observations
terraform_resources	AWS↔Terraform mapping
optimization_candidates	Deterministic findings
recommendations	AI and validated recommendation
terraform_plans	Plan metadata and JSON summary
approvals	Human approval decisions
verification_results	Post-change outcome
audit_logs	Immutable-ish event history

5. Resource Contract

resource_id
service
resource_type
account_id
region
configuration
tags
terraform_mapping
observed_at

6. Recommendation Contract

recommendation_id
resource_id
optimization_type
current_configuration
recommended_configuration
evidence
risk
confidence
estimated_monthly_savings
estimated_annual_savings
required_validation
status
created_at

7. API Design Principles

- Idempotent scans where practical
- Pagination
- Explicit timestamps
- Schema validation

- No secrets in responses
- Audit every state-changing operation
- Separate analysis from approval and execution